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Planned GLSI-100 (GP2 + GM-CSF) Phase III Clinical Trial, FLAMINGO-01

A Breakthrough Targeted Immunotherapy to Prevent Breast Cancer Recurrences

NASDAQ: **GLSI**

Snehal Patel, CEO

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Image: T-cells targeting cancer cell

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GLSI-100 (GP2 + GM-CSF) Executive Summary

Flamingo-01 - Current Phase III Trial with Interim Analysis: 9 amino acid HER2/*neu* peptide + GM-CSF immunotherapy for breast cancer in HER2/*neu* 3+, HLA-A2 patients following neoadjuvant, surgery, & adjuvant Herceptin or Kadcyla, led by Baylor & consortium of prominent networks - up to 150 sites in US & EU

- Conservative design of Phase III trial to reproduce Phase IIb results
- Phase IIb Trial Results: Randomized, multi-center (16 centers), placebo-controlled, substantial reduction in recurrence rate, peak immunity after 6 months, minimal to no side effects, no SAEs attributable to GP2, led by MD Anderson Cancer Center
- Potential Opportunities to Expand Market:
  - HER2/*neu* 1-2+ patients with Herceptin - increase market from 25% to 75%
  - ✓ Other HLA types – increase from 40-50% up to 80-100% of all patients
  - Combination with CD4/CD8 peptides and checkpoints
  - Other HER2/*neu* cancers
- NASDAQ Ticker "GLSI": Raised $40m since IPO

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First Year Share Price Performance After Presenting Key Phase IIb Data in 2020/2021

- Raised approximately $36.5M from IPO & Follow-on in 2020
- Peak share prices and trading volume driven by data publications
- 30x or 2,940% return from intraday high on 12/9/20, possibly highest intraday return for a non-penny stock from prior day close
- Added to Russell 2000 twice: 2021 and 2024
- Management & Directors ownership (tightly held): 75-85%
- Low float - Locked up from IPO in Sept 2020 through April 2026 at BOD discretion

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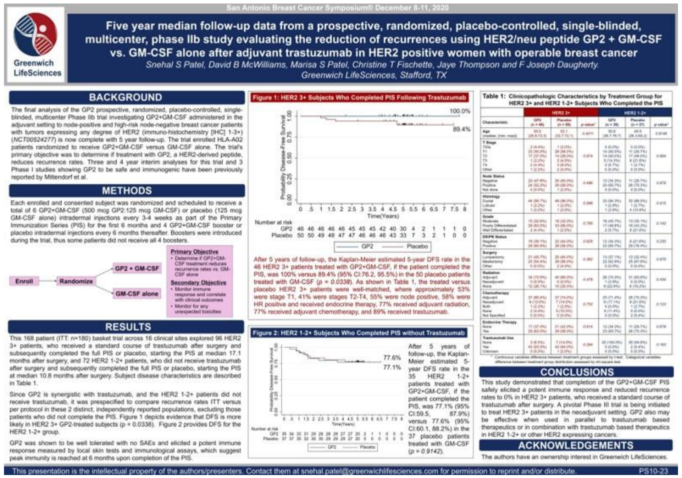
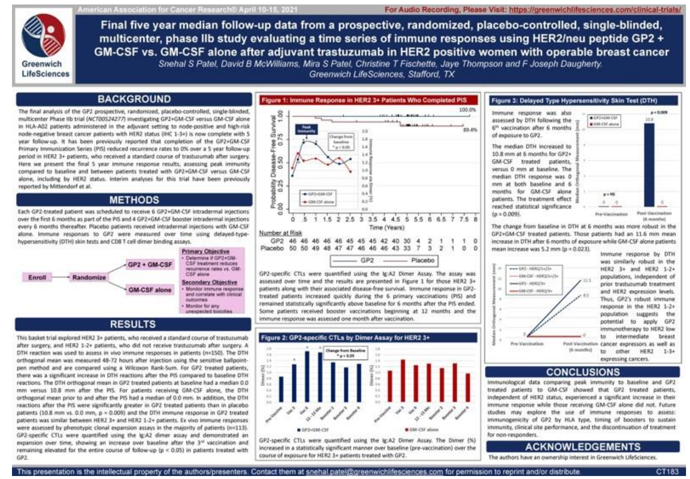
2020-2021 - Four Posters Highlighting Efficacy, Mechanism of Action, Safety, & Prognostic Factors

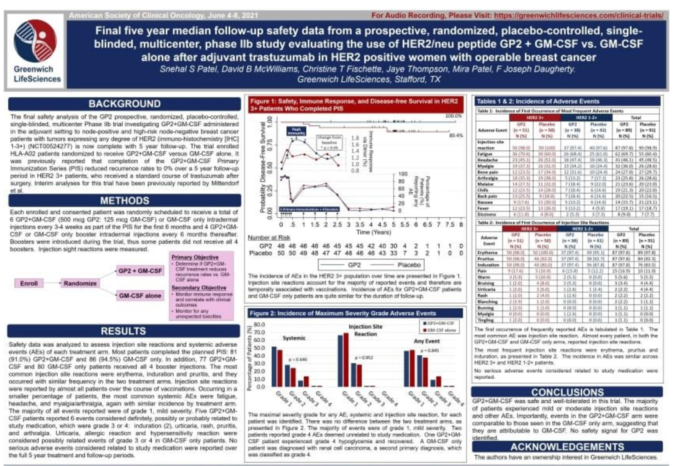
2020 San Antonio Breast Cancer Symposium (SABCS) 0% Recurrences Over 5 Years in Phase IIb Trial



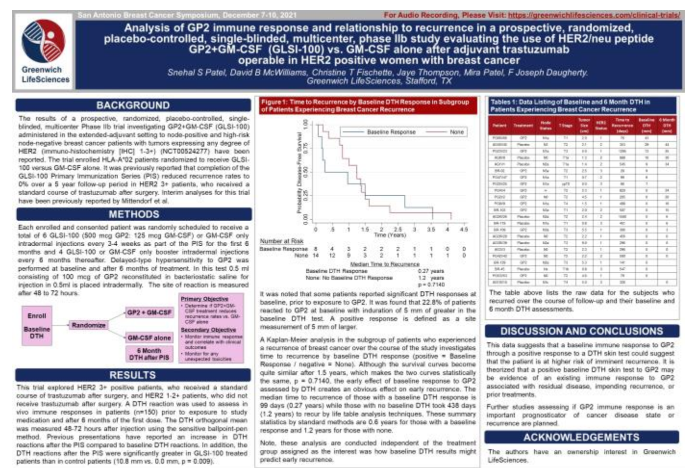
2021 American Association for Cancer Research (AACR) Immune Response Supports Mechanism of Action



2021 American Society of Clinical Oncology (ASCO) Injection Related Immune Reactions, No GP2 Related SAEs



2021 SABCS Baseline GP2 Immune Response May Predict Faster & Earlier Recurrence



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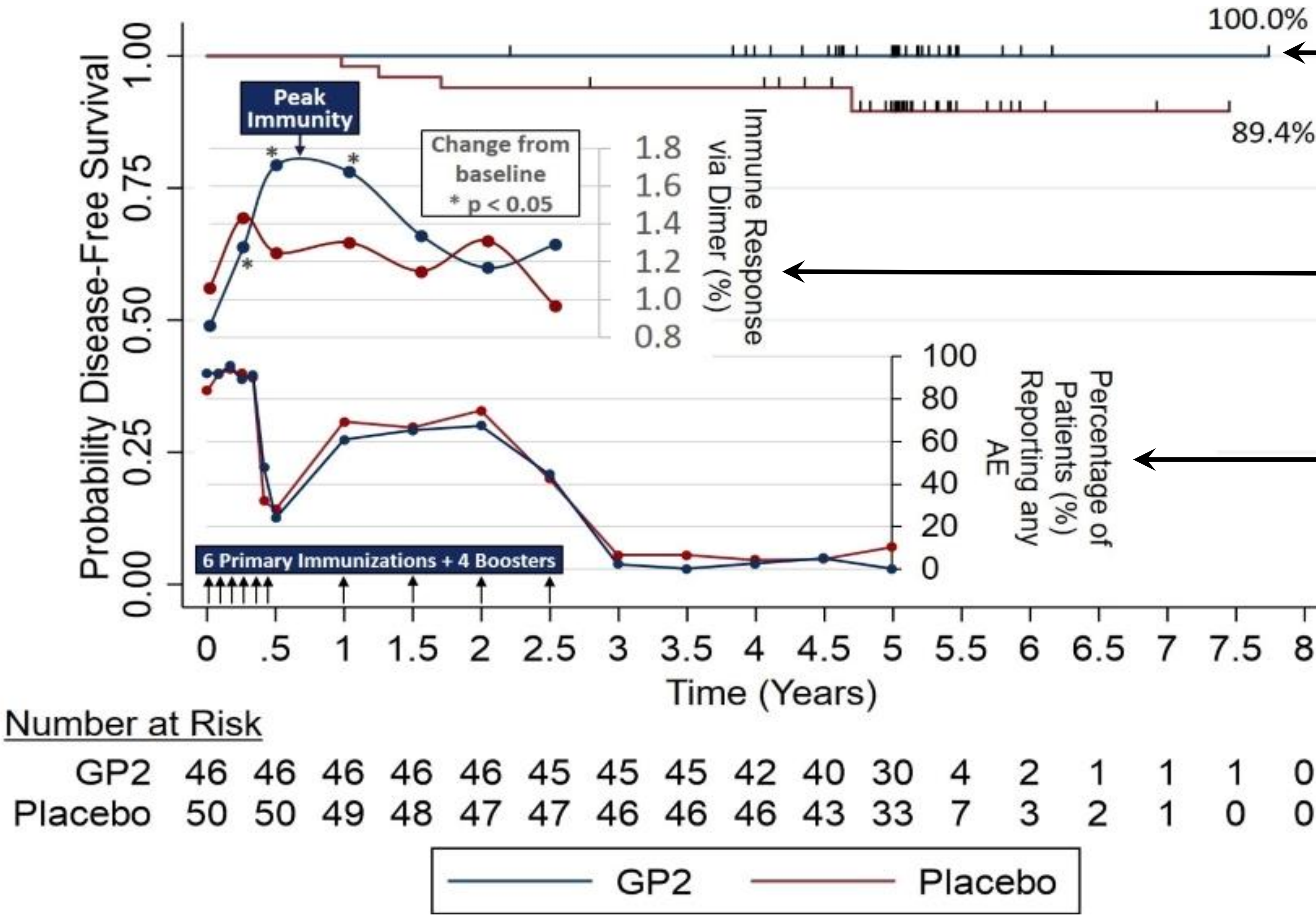
5 Year Data Set of GP2 Phase IIb Trial is Complete

HER2 3+ (Positive) Patients who Completed the Primary Immunization Series (PIS)



100% Disease Free Survival, 0% Metastatic Breast Cancer Recurrences**

Peak Immunity After 6 Months & PIS

Injection Site & Systemic Adverse Events (AE) are Positive Sign of Immune Response – No SAEs attributable to GP2 treatment

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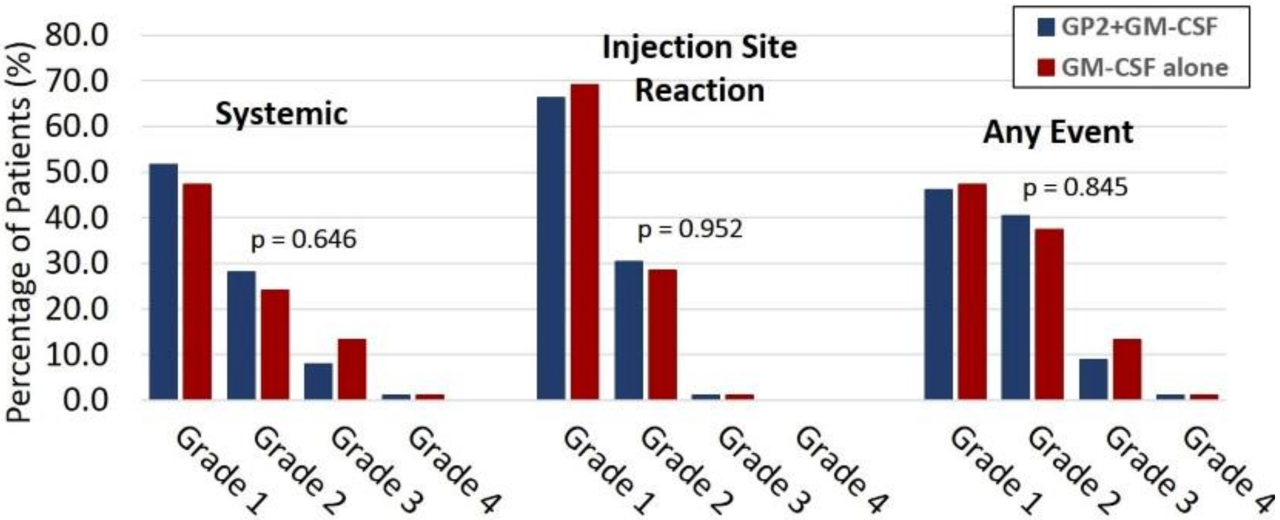
**Results subject to final Clinical Trial Study Report being prepared for filing of BLA per 10K, ending 12/31/23, & per Phase IIb trial tab of Company website.

ASCO 2021 – Systemic & Injection Site Reactions

Figure 2: Incidence of Maximum Severity Grade Adverse Events



The maximal severity grade for any AE, systemic and injection site reaction, for each patient was identified. There was no difference between the two treatment arms, as presented in Figure 2. The majority of events were of grade 1, mild severity. Two patients reported grade 4 AEs deemed unrelated to study medication. One GP2+GM-CSF patient experienced grade 4 hypoglycemia and recovered. A GM-CSF only patient was diagnosed with renal cell carcinoma, a second primary diagnosis, which was classified as grade 4.

No serious adverse events considered related to study medication were reported

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ASCO 2021 – Incidence of Adverse Events

Tables 1 & 2: Incidence of Adverse Events

Table 1: Incidence of First Occurrence of Most Frequent Adverse Events

	HER2 3+		HER2 1-2+		Total	
Adverse Event	GP2 (n = 51) N (%)	Placebo (n = 50) N (%)	GP2 (n = 38) N (%)	Placebo (n = 41) N (%)	GP2 (n = 89) N (%)	Placebo (n = 91) N (%)
Injection site reaction	50 (98.0)	50 (100)	37 (97.4)	40 (97.6)	87 (97.8)	90 (98.9)
Fatigue	36 (70.6)	30 (60.0)	26 (68.4)	25 (61.0)	62 (69.7)	55 (60.4)
Headache	23 (45.1)	26 (52.0)	1847.4)	19 (46.3)	41 (46.1)	45 (49.5)
Myalgia	19 (37.3)	16 (32.0)	13 (34.2)	10 (24.4)	32 (36.0)	26 (28.6)
Bone pain	12 (23.5)	17 (34.0)	12 (31.6)	10 (24.4)	24 (27.0)	27 (29.7)
Arthralgia	18 (35.3)	19 (38.0)	5 (13.2)	7 (17.1)	23 (25.8)	26 (28.6)
Malaise	14 (27.5)	11 (22.0)	7 (18.4)	9 (22.0)	21 (23.6)	20 (22.0)
Chills	12 (23.5)	14 (28.0)	7 (18.4)	6 (14.6)	19 (21.3)	20 (22.0)
Back pain	13 (25.5)	9 (18.0)	7 (18.4)	6 (14.6)	20 (22.5)	15 (16.5)
Nausea	9 (17.6)	15 (30.0)	5 (13.2)	6 (14.6)	14 (15.7)	21 (23.1)
Fever	12 (23.5)	13 (26.0)	5 (13.2)	4 (9.8)	17 (19.1)	17 (18.7)
Dizziness	6 (11.8)	4 (8.0)	2 (5.3)	3 (7.3)	8 (9.0)	7 (7.7)

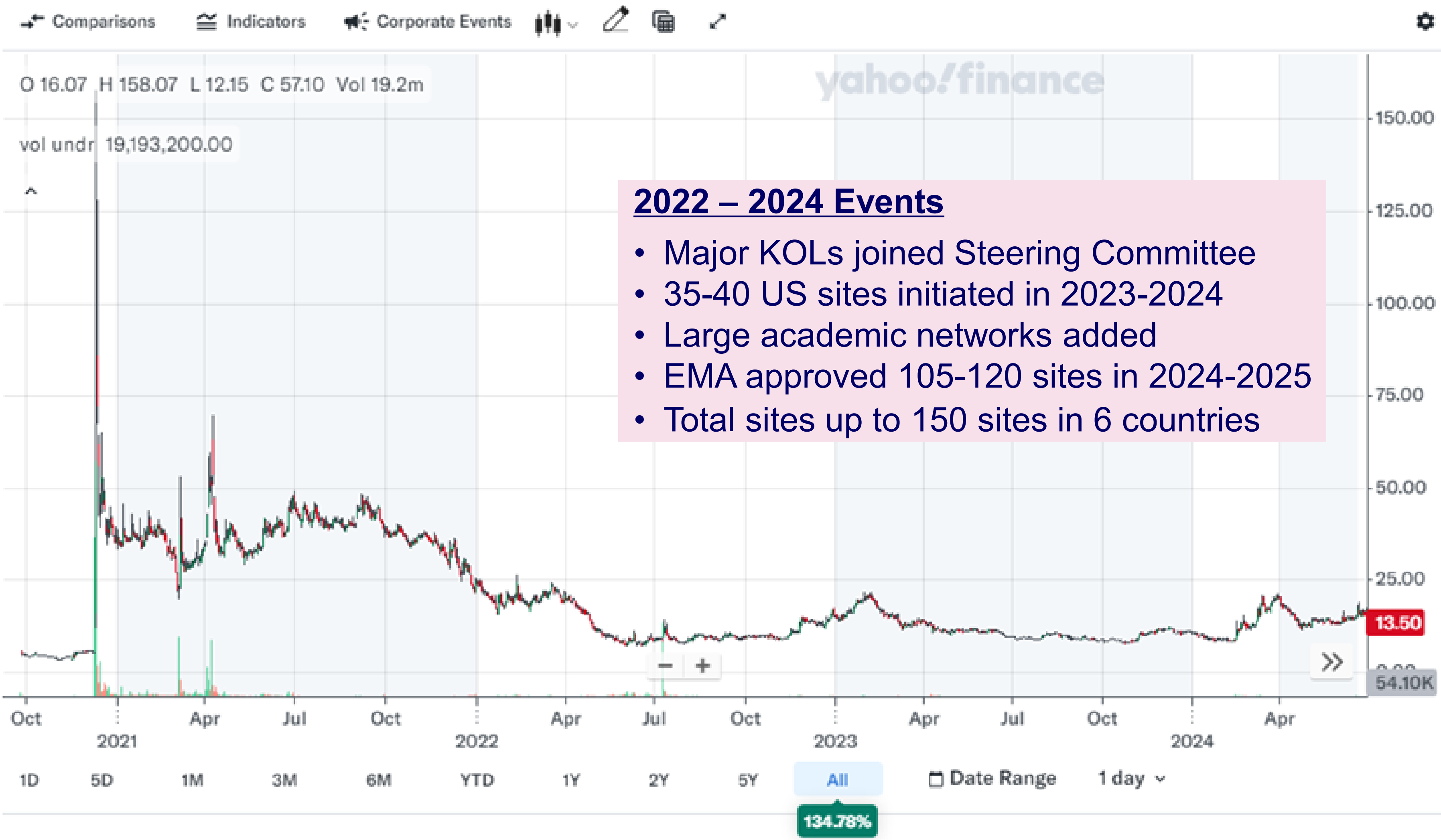
Table 2: Incidence of First Occurrence of Injection Site Reactions

	HER2 3+		HER2 1-2+		Total	
Adverse Event	GP2 (n = 51) N (%)	Placebo (n = 50) N (%)	GP2 (n = 38) N (%)	Placebo (n = 41) N (%)	GP2 (n = 89) N (%)	Placebo (n = 91) N (%)
Erythema	50 (98.0)	50 (100.0)	37 (97.4)	39 (95.1)	87 (97.8)	89 (97.8)
Pruritus	50 (98.0)	46 (92.0)	37 (97.4)	38 (92.7)	87 (97.8)	84 (92.3)
Induration	50 (98.0)	40 (80.0)	37 (97.4)	36 (87.8)	87 (97.8)	76 (83.5)
Pain	9 (17.6)	5 (10.0)	6 (15.8)	5 (12.2)	15 (16.9)	10 (11.0)
Warm	3 (5.9)	5 (10.0)	2 (5.3)	0 (0.0)	5 (5.6)	5 (5.5)
Bruising	1 (2.0)	4 (8.0)	2 (5.3)	0 (0.0)	3 (3.4)	4 (4.4)
Urticaria	1 (2.0)	3 (6.0)	1 (2.6)	1 (2.4)	2 (2.2)	4 (4.4)
Rash	1 (2.0)	2 (4.0)	1 (2.6)	0 (0.0)	2 (2.2)	2 (2.2)
Blanching	2 (3.9)	1 (2.0)	0 (0.0)	0 (0.0)	2 (2.2)	1 (1.1)
Burning	1 (2.0)	1 (2.0)	0 (0.0)	0 (0.0)	1 (1.1)	1 (1.1)
Myalgia	0 (0.0)	0 (0.0)	1 (2.6)	0 (0.0)	1 (1.1)	0 (0.0)
Tingling	1 (2.0)	0 (0.0)	0 (0.0)	0 (0.0)	1 (1.1)	0 (0.0)

The first occurrence of frequently reported AEs is tabulated in Table 1. The most common AE was injection site reaction. Almost every patient, in both the GP2+GM-CSF and GM-CSF only arms, reported injection site reactions.

The most frequent injection site reactions were erythema, pruritus and induration, as presented in Table 2. The incidence in AEs was similar across HER2 3+ and HER2 1-2+ patients.

4 Year Performance Since IPO – Planning & Initiation of Phase III Trial in up to 150 sites in 6 Countries



2022 – 2024 Events

- Major KOLs joined Steering Committee
- 35-40 US sites initiated in 2023-2024
- Large academic networks added
- EMA approved 105-120 sites in 2024-2025
- Total sites up to 150 sites in 6 countries

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Flamingo-01 Steering Committee

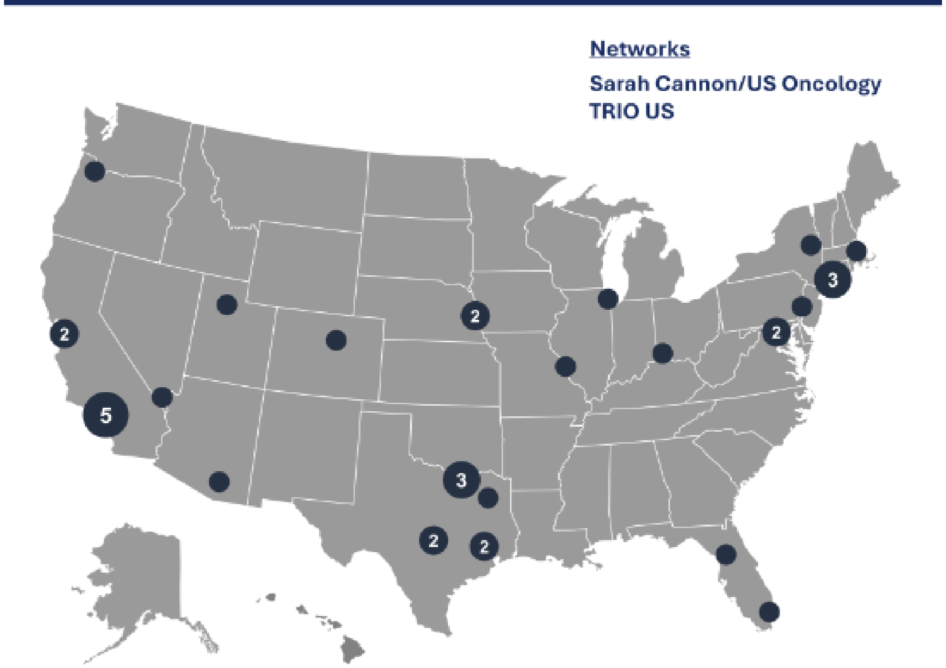
The Steering Committee is comprised of the following members :

- Dr. Mothaffar F. Rimawi – Professor of Medicine at the Baylor College of Medicine
- Dr. Francois-Clement Bidard – Professor of Medical Oncology, UVSQ/Paris Saclay University, Head of Breast Cancer Group, Institut Curie (Unicancer)
- Dr. William J. Gradishar – Professor of Medicine at the Feinberg School of Medicine at Northwestern University
- Dr. Sibylle Loibl – Professor (apl) Goethe University Frankfurt/M, Clinical Consultant Centre for Haematology and Oncology/Bethanien Frankfurt/M (GBG)
- Dr. Miguel Martin – Professor of Medicine, Head, Medical Oncology Service, Gregorio Marañón General University Hospital, Complutense University, Madrid, GEICAM
- Dr. Joyce A. O'Shaughnessy – Celebrating Women Chair in Breast Cancer, Baylor University Medical Center & Chair, Breast Cancer Program, US Oncology/Sarah Cannon
- Dr. Hope S. Rugo – Professor of Medicine and Winterhof Family Professor of Breast Oncology, University of California, San Francisco
- Dr. Laura M. Spring – Assistant Professor, Medicine, Harvard Medical School, Attending Physician, Medical Oncology, Massachusetts General Hospital
- Dr. Cesar A. Santa-Maria – Associate Professor of Oncology, Breast and Gynecological Malignancies Group, Director of Breast Cancer Trials, Johns Hopkins SKCCC

Flamingo-01 Clinical Site Map – Up to 150 sites in US/EU

Participating networks: USOncology/Sarah Cannon, TRIO-US, GEICAM (Spain), Unicancer (France), GBG (Germany), GIM (Italy)

US Sites	EU Sites



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Breast Cancer – *Still* a Substantial Unmet Need

- ***Unmet Need is to address the 50% of recurring patients who do not respond to Herceptin or Kadcyla – an opportunity for GP2.***
- ***Adjuvant Setting: Following breast cancer surgery, HER2/neu 3+ patients receive Herceptin in the first year and then hope that their breast cancer will not recur, with the odds of recurrence slowly decreasing over the first 5 years. Herceptin reduces recurrence rates from 25% to 12%.***
- ***Neoadjuvant Setting: Kadcyla is approved for use in patients with residual disease, who do not achieve a pCR, as determined at time of surgery. Kadcyla reduces recurrence rates from 22% to 11%.***
- ***Neither Perjeta or Nerlynx fully address this unmet need, even in their most efficacious subpopulations.***

GP2 Addresses Unmet Need: GP2 & GM-CSF starting in Year 2 act with minimal to no side effects & no SAEs.

In the initial GP2 indication in the US & EU, ~44k new patients could be treated per year, saving up to 4,000 to 5,000 lives per year.
(6.25% * 700k new pts = 44k pts/yr for HLA-A*02)

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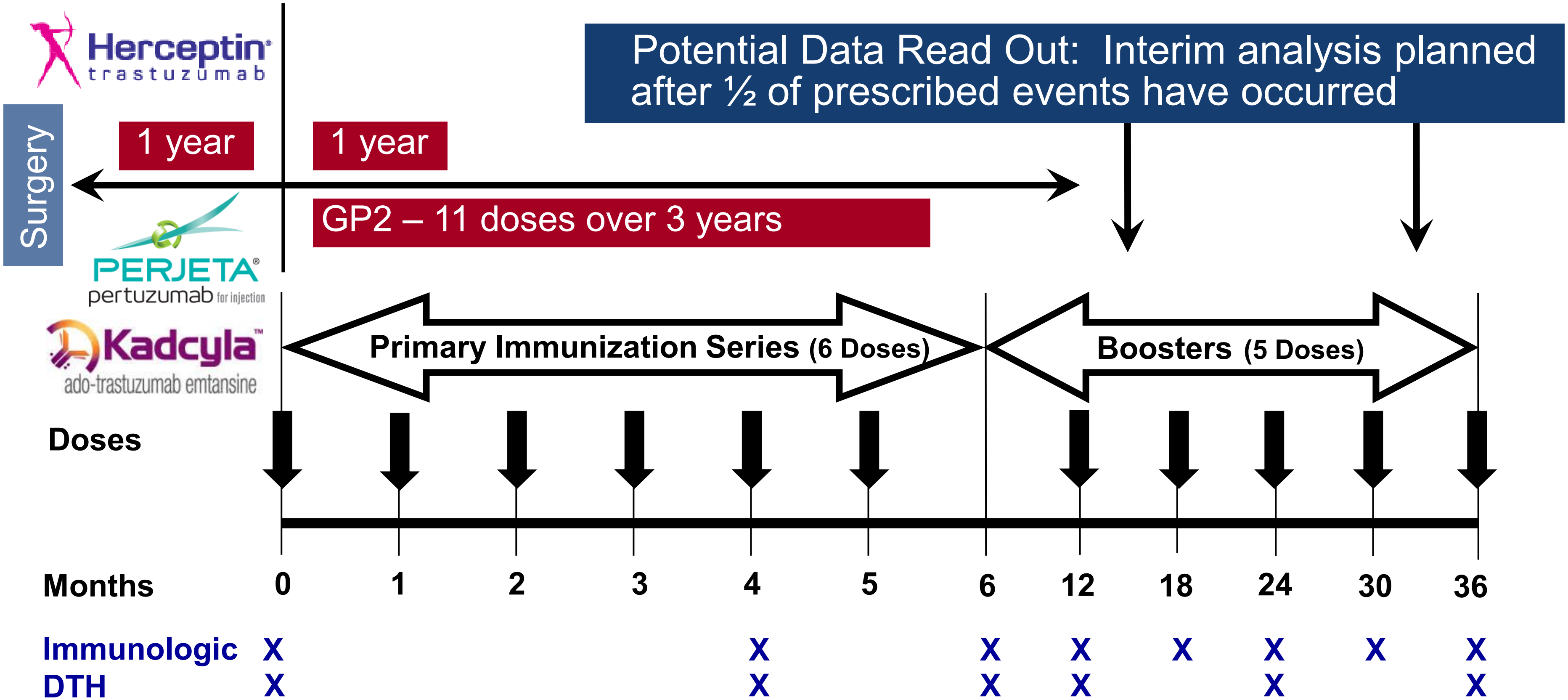
pCR = pathologic complete response, the lack of all signs of cancer in tissue samples remove during surgery or biopsy due to Neoadjuvant treatment.

GP2 Phase III Clinical Trial Dosing



- Study allows prior use of pertuzumab, trastuzumab, and ado-trastuzumab emtansine and concurrent neratinib
- Final DTH/immunologic assays at 48 months and at time of recurrence

GP2 Market Positioning & Feedback from KOLs

- As only injection site reactions were observed (which speaks to the immunogenicity of GP2) and with no SAEs attributable to GP2, GP2 can be positioned as the final treatment for patients post surgery
- Patients are seeking a de-escalation and a return to normal life free of toxic treatments, especially if the chance of recurrence is reduced substantially
- GP2 can be the treatment that will naturally overlap with or follow Herceptin, Kadcyla, or Enhertu or any of the other Herceptin derivatives being developed

5% Metastatic

11 GP2 Doses Over 3 Years

Preventing Recurrences may prevent 95% of Metastatic Breast Cancer

Diagnosis	Primary Treatment	Follow-up / Surveillance Period	Recurrence
Imaging *Biopsy*	*Surgery* *Chemotherapy* *Radiation* *Anti-HER2 Rx*	*Anti-Estrogen Rx* *Observation* *Cannot Predict Recurrence* *Patient Anxiety*	*Imaging* *Labs* *Anti-HER2 Rx* *Chemotherapy*

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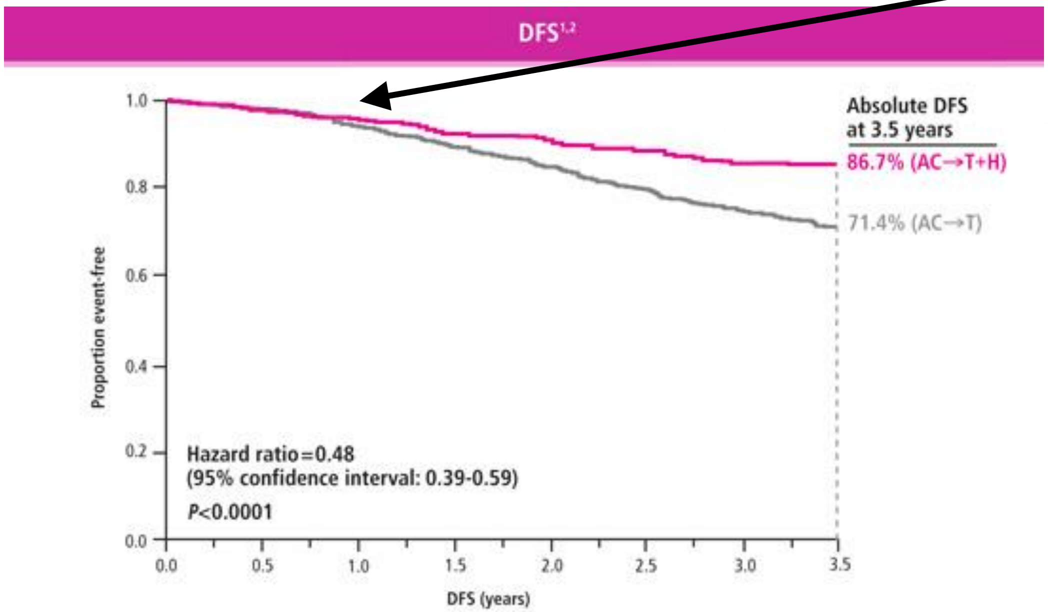
Synergy with Herceptin Alone **

** 5 Year Disease Free Survival without use of Kadcyla, Perjeta, Nerlynx, Enhertu, or Tukysa

Herceptin Approved for Adjuvant Treatment of HER2/*neu* 3+ Breast Cancer

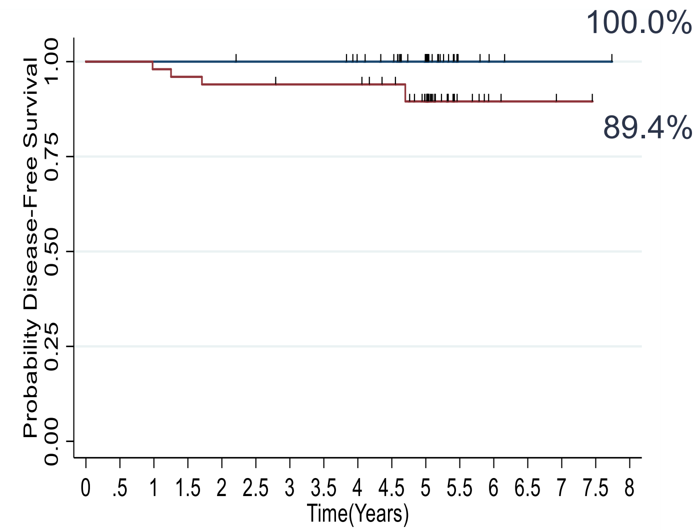


Number at risk (N= 3752)

AC→T+H	1872	1529	1240	997	764	575	426	239
AC→T	1880	1490	1159	926	689	534	375	195

AC=doxorubicin/cyclophosphamide T=paclitaxel H=Herceptin

Phase IIb Results for GP2 Target Population, if Fully Immunized (median 5 years follow-up)



Number at risk

	0	.5	1	1.5	2	2.5	3	3.5	4	4.5	5	5.5	6	6.5	7	7.5	8
GP2	46	46	46	46	46	45	45	45	42	40	30	4	2	1	1	1	0
Placebo	50	50	49	48	47	47	46	46	46	43	33	7	3	2	1	0	0

GP2 Placebo

Joint Analysis Trial

*DFS – Disease Free Survival

ROW strategy: GP2 could eventually be developed with only trastuzumab biosimilar

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GP2 Clinical Data:

GP2 is Immunogenic & Clinically Effective

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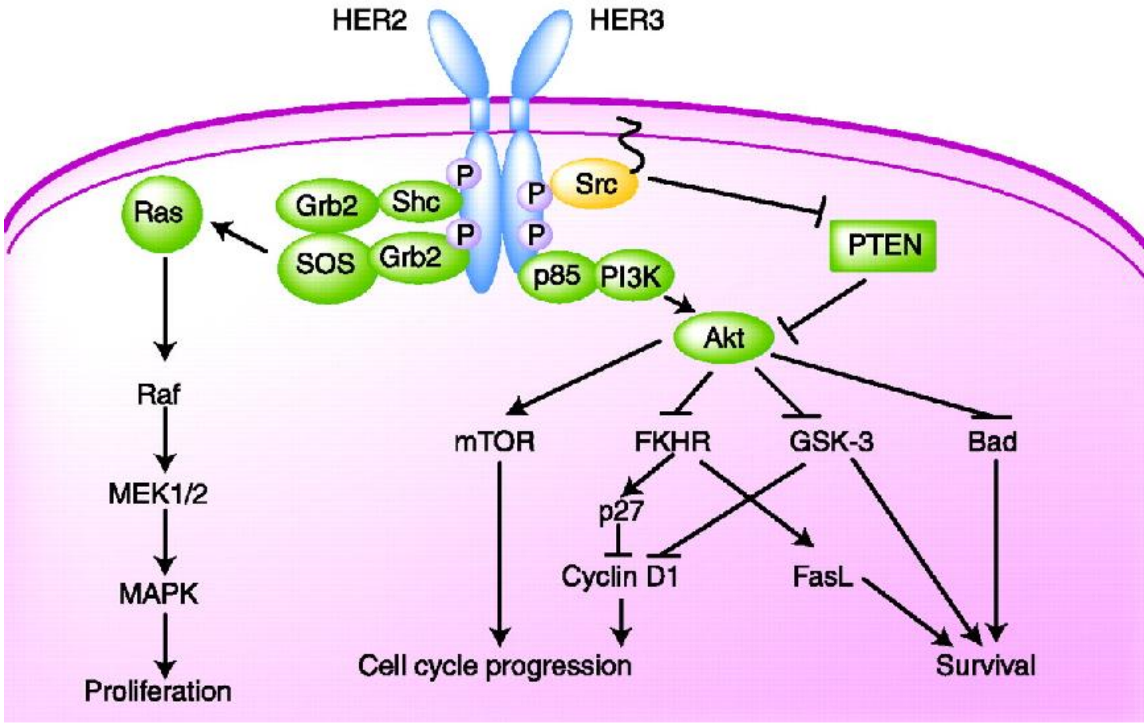
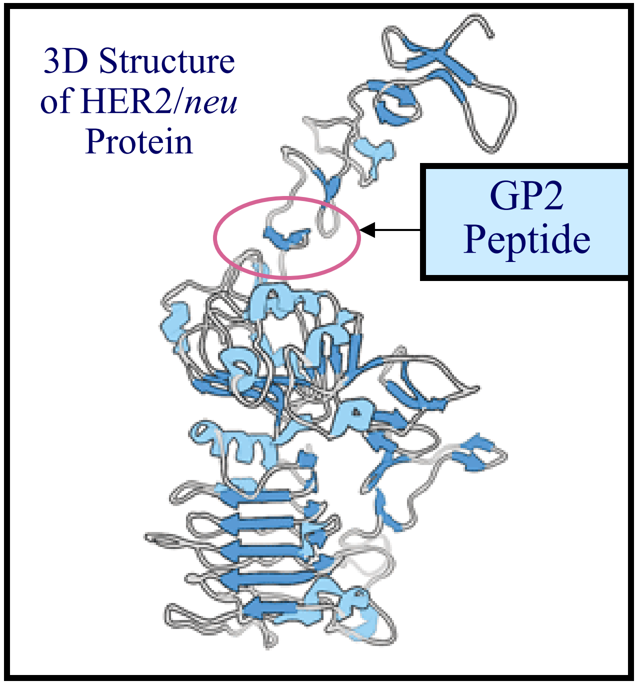
HER2/*neu* Signaling Pathway Well Studied

- **HER2/*neu* pathway activates cancer cell proliferation**
- **Overexpression of HER2/*neu* correlates strongly with aggressive cancers**



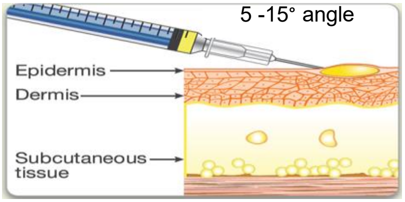


GP2 Product Description & Mechanism of Action

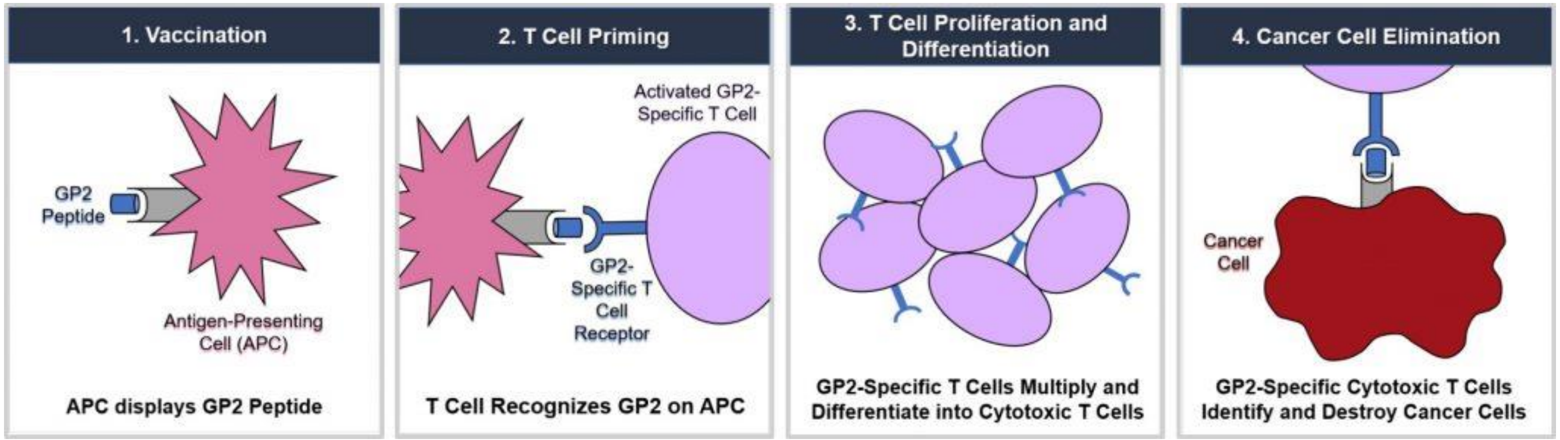
- 9 amino acid transmembrane peptide segment of HER2/*neu* protein
- Intradermal injection in combination with an FDA-approved immunoadjuvant GM-CSF, following 1st year of Herceptin treatment in Adjuvant Setting

Leukine® sargramostim

Herceptin® trastuzumab



- Given once per month for six months followed by 5 booster doses every 6 months = 11 doses over 3 years
- <u>Mechanism of Action</u>: 4 primary steps, followed by a secondary epitope spreading & broader immune response



Summary of GP2 Completed Trials - N=146 GP2-Treated Patients to Date with No SAEs Attributable to GP2

Study	Design and Control	Product, Dose, and Route	Regimen	Number of Subjects	Population	Duration of Follow-Up
Phase 1b 04-20017, (MCHL-SG (40-38a))	3x3 Dose-escalation	• GP2 at 100, 500, 1000mcg • GM-CSF at 250mcg (reduced to 125mcg in many subjects) • Intradermal	6 doses, every 3-4 weeks	18	• Breast cancer • HER2/neu 1-3+ • HLA-A*02 • Node negative	Primary safety follow-up for the duration of treatment + 30 days.
Phase 1b (C.2008.146)	3x3 Dose-escalation	• GP2+GM-CSF • GP2 at 100, 500, 1,000mcg • GM-CSF at 125, 250mcg • Intradermal • Concurrent iv trastuzumab	6 doses, every 3 weeks	17	• Breast cancer • HER2/neu 1-3+ • HLA-A*02 and HLA-A*03	Primary safety follow-up for the duration of treatment + 30 days.
Phase 1	3x3 Dose-escalation	• GP2+AE37+GM-CSF • GP2 at 100, 250, 500mcg • AE37 at 100, 250, 500mcg • GM-CSF at 125mcg • Intradermal	6 doses, 1 month apart	22	• Breast and ovarian cancer • HER2/neu 1-3+ • HLA-A*02 and HLA-A*03	1.5 years
Phase 2b (C.2007.098)	Randomized, Single-Blind	• GLSI-100 or GM-CSF alone • GP2 500mcg • GM-CSF 125mcg	• 6 doses, 1 month apart • 4 boosters beginning at 12 mo. then every 6 mo.	180 GLSI-100 (n = 89) GM-CSF alone (n = 91)	• Breast cancer • HER2/neu 1-3+ • HLA-A*02 • Node-positive and High-risk node-negative	5 years

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GP2 Phase III Trial Commenced:

Strategy – Conservatively Reproduce Phase IIb Trial in Larger Population

Flamingo-01 - Phase III Trial Flamingo-01 Has Commenced

- Approximately 120 sites are active, with another 30 approved EU sites in start-up
- Flamingo-01 is evaluating the safety and efficacy of GLSI-100 (GP2 + GM-CSF) in HER2/neu positive breast cancer patients who had residual disease or high-risk pathologic complete response at surgery
- What is next – transition into pre-commercialization activities:
  - Working with the FDA in preparation for a BLA submission and commercial launch
  - Implementing a global strategy for launching GP2 in international markets outside the US and Europe
  - 1st commercial lot completed - large scale manufacturing, packaging, and marketing
- Additional activities/milestones:
  - Phase III clinical trial progress and open label data may be presented at major conferences
  - Licensing discussions may accelerate as the interim analysis approaches
  - Other assets may be developed by acquisition or internal research, including T cell therapies that may be discovered in the Phase III trial by studying GP2's robust immunogenicity
  - Additional patents for GP2 based on the Phase III trial findings, manufacturing, and pharmacy procedures are planned to be filed to extend patent life

Flamingo-01 - Phase III Trial Schema & Interim Analysis

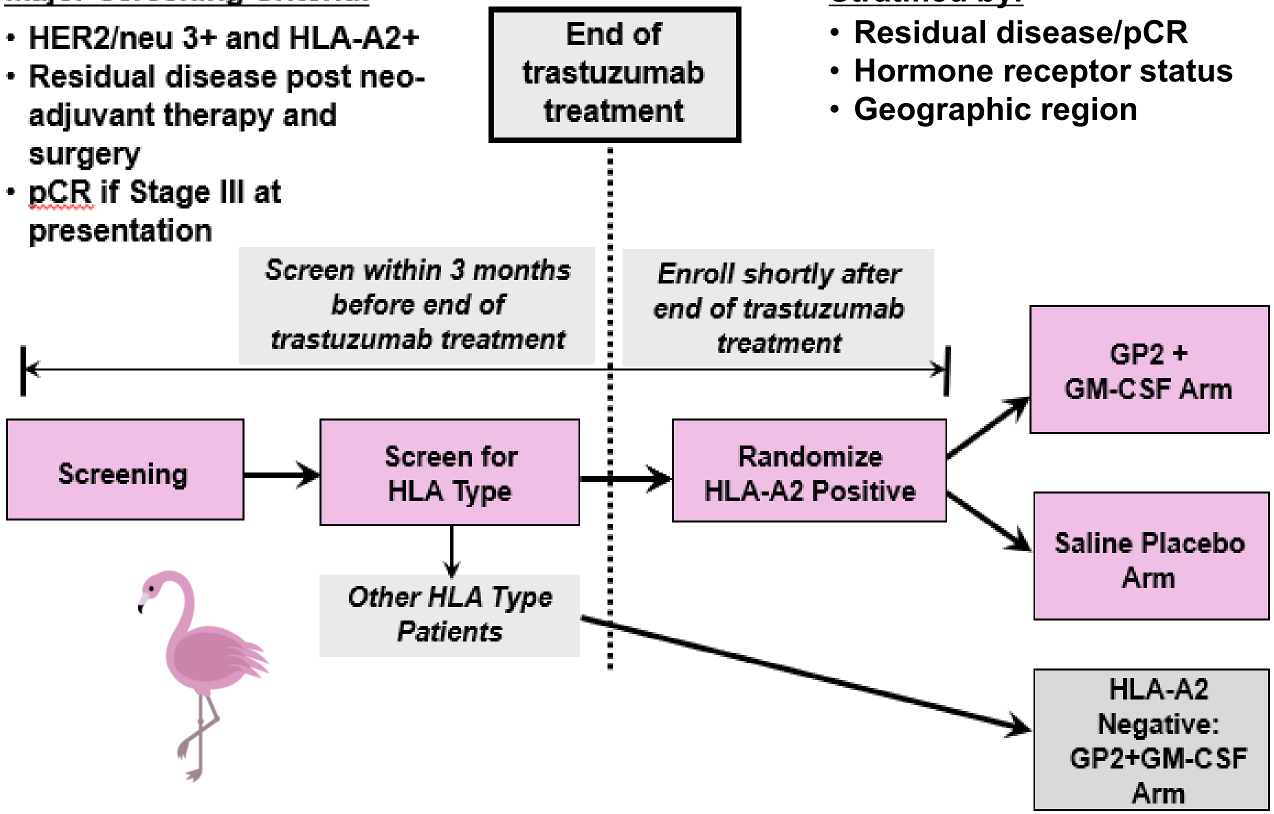
Phase III Trial Schema

Major Screening Criteria:

- HER2/neu 3+ and HLA-A2+
- Residual disease post neo-adjuvant therapy and surgery
- pCR if Stage III at presentation

Stratified by:

- Residual disease/pCR
- Hormone receptor status
- Geographic region



Interim Analysis Design

Preliminary Sizing of Trial:

Approximately 498 subjects will be enrolled. To detect a hazard ratio of 0.3 in IDFS, 28 events will be required. An interim analysis for superiority and futility will be conducted when at least half of those events, 14, have occurred. This sample size provides 80% power if the annual rate of events in placebo-treated subjects is 2.4% or greater.

Considering continuation of enrollment until an interim analysis is conducted and the appropriate size of each arm can be further assessed.

- New preliminary positive immune response data in non-HLA-A*02 open label third arm could lead to effectively a second Phase III trial
- Potential for multiple pathways for marketing approval of GLSI-100 and a doubling of market

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Flamingo-01 - HLA Prevalence, Safety, & Immune Response Data is Trending as Expected in Both HLA Arms

Preliminary Flamingo-01 data available to date is summarized in a blinded manner. This summary may not reflect results at the end of the study.

- Safety data in both the HLA-A*02 treated and placebo arms and the third open label arm with all other HLA types, shows that GP2 immunotherapy continues to be well-tolerated and that no safety signal for GP2 has yet to be identified across all arms of the study. Like the Phase IIb clinical trial, the most frequent adverse event is injection site reaction, which is also a sign of an immune response
- Immune response data in both the HLA-A*02 treated and placebo arms and the third open label arm with all other HLA types (non-HLA-A*02), show that GLSI-100 is creating an immune response over time as the frequency of patients exhibiting an injection site reaction or GP2 Delayed-Type Hypersensitivity (DTH) skin test reaction as measured by induration or erythema is increasing with increased vaccinations
  - The number of patients experiencing an immune response increased over time from baseline through the 4th to 6th month vaccinations in both HLA-A*02 and non-HLA-A*02 arms
  - In addition, a baseline immune response to GP2 is being observed before any treatment with GP2 in both HLA-A*02 and non-HLA-A*02 arms. This result suggests that GP2 is a natural antigen, that may have been part of an immune response in the patient during prior trastuzumab or other treatments
  - These two general immune response findings, immune response at baseline and increasing immune response over time, were also observed in the Phase IIb clinical trial for HLA-A*02 only patients
  - The non-HLA-A02 types that are most commonly being enrolled in FLAMINGO-01 include HLA-A*03, HLA-A*24, HLA-A*01, HLA-A*11, HLA-A*68, HLA-A*29, HLA-A*30, HLA-A*23, and HLA-A*33.

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Expand by HLA - GP2-HLA Observed & Predicted Prevalence

Preliminary Flamingo-01 data available to date is summarized in a blinded manner. This summary may not reflect results at the end of the study.

- Across all screened patients, HLA-A*02 prevalence is about 46%. The double HLA-A*02 prevalence, in patients who have received HLA-A*02 alleles from both parents, is about 8%. The HLA-A*03, HLA-A*24, and HLA-A*01 prevalences are about 20-25% for each allele. The HLA-A*11, HLA-A*68, HLA-A*29, and HLA-A*30 prevalences are about 9-12% for each allele.
- In those screened patients who self-report race as:
  - White - HLA-A*02 genes are prevalent in 50% of the patients
  - Hispanic or Latino - HLA-A*02 alleles are prevalent in 50% of the patient
  - Black or African-American - HLA-A*02 alleles are prevalent in 40% of the patients
  - Asian - HLA-A*02 alleles are prevalent in 17% of the patients

Frequency of Class I MHC alleles HLA in the North America - Data from HLA Matchmaker

Allele	White	Allele	Black	Allele	Hispanic	Allele	Asian/ Pacific
A*02:01	45.6%	C*04:01	29.0%	A*02:01	37.1%	A*11:01	38.4%
C*07:01	27.7%	C*07:01	25.4%	C*04:01	25.4%	A*24:02	33.7%
A*01:01	27.4%	C*06:02	23.0%	A*24:02	24.9%	C*07:02	33.3%
A*03:01	23.8%	A*02:01	22.3%	C*07:02	24.2%	C*01:02	27.7%
C*07:02	21.5%	A*23:01	20.7%	C*07:01	20.8%	A*33:03	23.3%
C*04:01	21.2%	C*02:02	19.0%	C*03:04	14.4%	C*08:01	21.6%
B*44:02	20.2%	A*03:01	18.7%	A*03:01	14.3%	C*03:04	19.9%
B*07:02	18.1%	C*07:02	18.1%	B*07:02	13.2%	A*02:01	18.1%

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GP2 May Address Unmet Need in Both HER2/neu 3+ Adjuvant & Neoadjuvant Settings

Adjuvant

Surgery

ER/PR + 4+ LN	ER/PR – LN+
Chemotherapy + trastuzumab → neratinib (2y total)	Chemotherapy + trastuzumab + pertuzumab (1y total)

Neoadjuvant

ER/PR + Any LN	ER/PR – LN+
Chemotherapy + trastuzumab	Chemotherapy + trastuzumab + pertuzumab

Surgery

pCR trastuzumab	*Residual disease* *TDM1*	*pCR* trastuzumab	*Residual disease* *TDM1*

Projected 5 Year Recurrence Rates

9-12%	9-12%	5-10%	17%	5-10%	17%

Conclusion: GP2 should eventually be pursued in both settings for all HER2 positive patients

Target Population for Phase III Trial: Residual Disease & High Risk pCR

Annual recurrence rate = 3.0-3.4%

Annual recurrence rate design for Phase III trial = 2.4%

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GP2 Specific & Secondary Immune Response TCR Sequencing

Multiple Open Label Immune Response Assessments are Planned to Potentially Identify GP2 Specific T cells, Biomarkers, Booster Strategies, and Early Recurrers

1. Delayed Type Hypersensitive (DTH) skin test for in vivo immune response to GP2 (20% of dose)
2. Injection site reaction measurements which are correlated to DTH in Phase IIb trial
3. Single cell sequencing of T cells resulting from spiked dendritic cells
4. TCR sequencing from frozen blood samples over course of Phase III trial (Example: extensive sequencing of COVID-19 infection & vaccinated patient samples)
5. Potential for ctDNA analysis of plasma
6. Potential for Elispot assay for quantification of functional GP2 specific T cells

GP2 Commercial Opportunity

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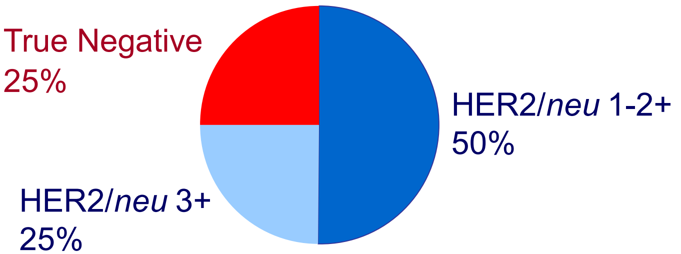
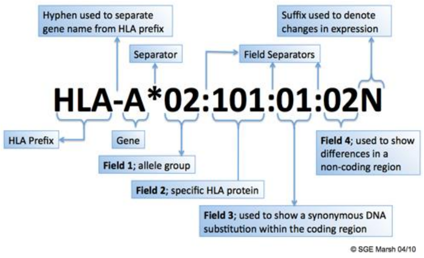
Potential Indications of GP2 & Herceptin in Various Populations in Neoadjuvant Setting

- HER2/*neu* 3+ protein over-expression (25%) & 1-2+ expression (50%)
  - All breast cancer patients are tested for HER2/*neu* expression by immunohistochemistry (IHC) or fluoresecence in situ hybridisation (FISH)



- Node Positive (60%) & High Risk Node Negative (40%)
  - Node positive – cancer has spread to lymph nodes
  - High risk node negative – no cancer in lymph nodes but at high risk for recurrence
  - The more lymph node involvement the more aggressive the cancer
- Hormone Receptor Positive (60%) & Hormone Receptor Negative (40%)
- HLA Type: HLA-A2 (40-50%) & HLA-A3,A24 (30%)
  - Human leukocyte antigen (HLA) presents peptide from inside cancer cell to killer T-cells
  - HLA also presents injected peptide to create killer T-cells following intradermal injection



- 30% of 700k new US and EU breast cancer patients per year could lead to up to 210k new patients per year for GP2 (6.25% * 700k new pts = 44k pts/yr for HLA-A*02)
- 30% of 9.5m long term US and EU breast cancer survivors could be candidates for GP2 (Sources: American Cancer Society, Economic Impact, & European Cancer Information System 2025)

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Commercial Opportunity for GP2 in Breast Cancer in US

- 1 in 8 U.S. women (12.8%) will develop invasive breast cancer over her lifetime, with 317k new breast cancer patients per year in 2025
- An estimated 42k female breast cancer deaths will occur in the US in 2025
- GP2's target market is 6-30% of available breast cancer market or up to 2.4x that of Herceptin in adjuvant setting
- GP2 could be a long term treatment that treats survivors (4m as of 2025)
- Herceptin/Perjeta/Nerlynx/Kadcyla pricing from \$75k - \$125k per patient per year
- 11 doses over 3 years in initial indication

	Herceptin	GP2
US Market Potential (Size = 4m current breast cancer survivors and 317k new patients per year)		
HER2/*neu* Expressors (1-3+)	25% (3+)	25-75% (1-3+)
HLA Type	100%	50-80% (2/3/24/26)
Residual Disease or High Risk pCR	50%	50%
Target Market Potential	12.5%	6.25 - 30%
Theoretical New Patients per Year	35,250	19,800 – 95,100 in US only
Adjuvant Patients Treated per Year (est. from sales)	27,000 – 40,000	44k - 210k pts/yr for both US & EU.
Estimated Adjuvant Setting US Revenue (\$ billions)	\$2-3	
Estimated Price (first year)	\$74,500	TBD (6 primary + 1 booster)
Estimated Price (booster)	Not Approved	TBD (4 boosters over 2 years)
Estimated 2017 Global Revenue (\$ billions)	\$7	
Adjuvant Setting	\$2-3	US Revenue could double when adding EU market.
Metastatic Breast Cancer	\$4-5	

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Potential Commercial Opportunities / Additional Indications for GP2: HER2/*neu* Expressed in Multiple Cancers

Figure 3. Leading Sites of New Cancer Cases and Deaths – 2021 Estimates

Estimated New Cases

Male		
Prostate	248,530	26%
Lung & bronchus	119,100	12%
Colon & rectum	79,520	8%
Urinary bladder	64,280	7%
Melanoma of the skin	62,260	6%
Kidney & renal pelvis	48,780	5%
Non-Hodgkin lymphoma	45,630	5%
Oral cavity & pharynx	38,800	4%
Leukemia	35,530	4%
Pancreas	31,950	3%
All sites	**970,250**	

Female		
Breast	281,550	30%
Lung & bronchus	116,660	13%
Colon & rectum	69,980	8%
Uterine corpus	66,570	7%
Melanoma of the skin	43,850	5%
Non-Hodgkin lymphoma	35,930	4%
Thyroid	32,130	3%
Pancreas	28,480	3%
Kidney & renal pelvis	27,300	3%
Leukemia	25,560	3%
All sites	**927,910**	

Estimated Deaths

Male		
Lung & bronchus	69,410	22%
Prostate	34,130	11%
Colon & rectum	28,520	9%
Pancreas	25,270	8%
Liver & intrahepatic bile duct	20,300	6%
Leukemia	13,900	4%
Esophagus	12,410	4%
Urinary bladder	12,260	4%
Non-Hodgkin lymphoma	12,170	4%
Brain & other nervous system	10,500	3%
All sites	**319,420**	

Female		
Lung & bronchus	62,470	22%
Breast	43,600	15%
Colon & rectum	24,460	8%
Pancreas	22,950	8%
Ovary	13,770	5%
Uterine corpus	12,940	4%
Liver & intrahepatic bile duct	9,930	3%
Leukemia	9,760	3%
Non-Hodgkin lymphoma	8,550	3%
Brain & other nervous system	8,100	3%
All sites	**289,150**	

Estimates are rounded to the nearest 10, and cases exclude basal cell and squamous cell skin cancers and in situ carcinoma except urinary bladder. Estimates do not include Puerto Rico or other US territories. Ranking is based on modeled projections and may differ from the most recent observed data.

Denotes cancers where HER2/*neu* over expression has been reported

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Veteran Management Team / Board

- David McWilliams, MBA – Chairman, Board
  - 40 years of start-up / CEO experience
  - CEO of 2 private and 3 public biotech companies
- Snehal Patel, MS, MBA – CEO, Board
  - 30 years of biopharma / Wall Street experience
  - Large pharma operations / management experience
- Joe Daugherty, M.D. – CMO, Board
  - 35 years of biopharma experience
  - Assisted over 20 public and private companies
- Jaye Thompson, Ph.D. – VP Clinical & Regulatory
  - 30 years of active involvement in over 200 clinical trials for drugs, biologics and devices
  - Founder of multiple CROs
- Christine Fischette, Ph.D. – VP Business Development
  - 30 years of big pharma R&D & commercialization
  - Business development / multiple licensing transactions
- Eric Rothe – Board & Founder of GLSI
- Ken Hallock – Board & Major Investor

Manufacturing / Regulatory / IP

- GP2 manufactured by straightforward amino acid chemistry
  - Manufactured by FDA-approved commercial facility with multiple back-up facilities
  - Commercial manufacturing commenced
  - 3 clinical lots followed by 3 commercial lots
  - GM-CSF is commercially available, along with Saline/WFI, which will all be sold independently
- Discussing potency assay / HLA companion diagnostic
- GP2 registered as biologic with CBER – 12 years exclusivity in US
- GP2 issued patents provide protection through 2032 in the major markets (US, EU, Canada, Australia, & Japan), including ongoing prosecution in emerging markets - patent term extensions possible
- New patent applications

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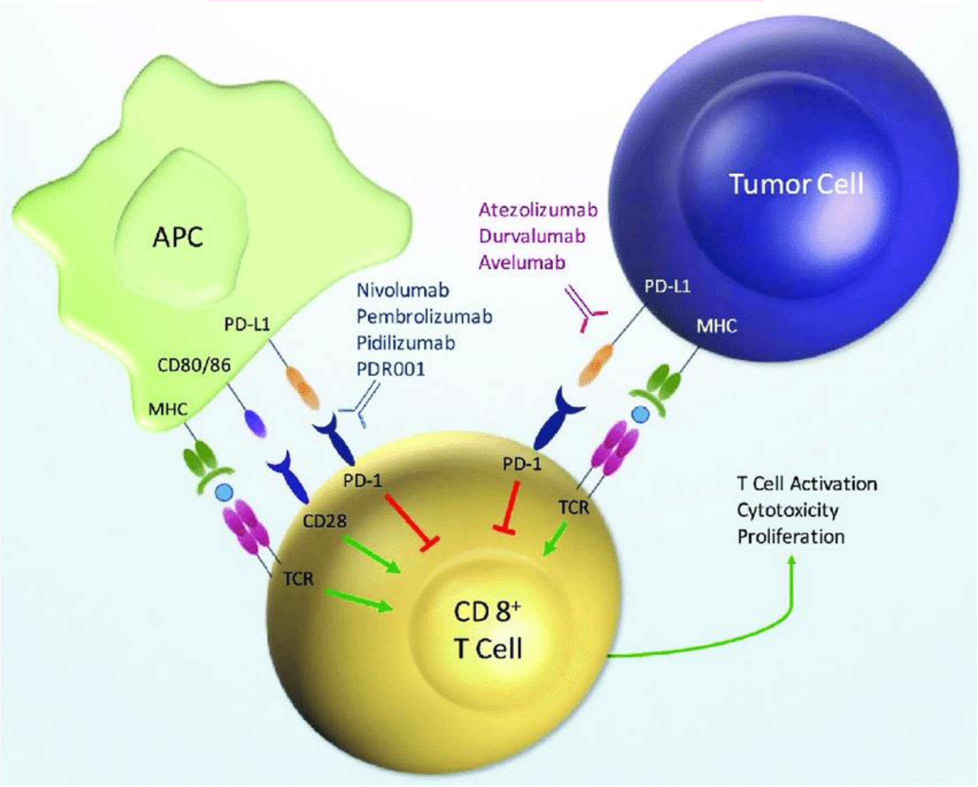
GLSI Strategy is to Conduct Additional GP2 Trials

Greenwich's current strategy is as follows:

- **Reproduce** Phase IIb trial in Phase III trial in HER2 positive patients only – no material changes to treatment regimen, upgrade immune response assays, expand to multiple HLA types
- **Optimize** GP2 treatment by starting treatment in neoadjuvant setting in another Phase II/III trial and utilize immune response data, if possible, to "optimize" timing of inoculations
- **Expand** to HER2 low breast cancer and other HER2 expressing cancers using optimized treatment methods and add checkpoint inhibitors

Add Checkpoint Inhibitors



ROE of developing GP2 could be high!

GP2 Conclusions: A Breakthrough Targeted Immunotherapy for Prevention of HER2/*neu* Cancer

- Flamingo-01 - Phase III Trial with Interim Analysis: 9 amino acid HER2/*neu* peptide + GM-CSF immunotherapy for breast cancer in HER2/*neu* 3+, HLA-A2 patients following neoadjuvant, surgery, & adjuvant Herceptin or Kadcyla, led by Baylor & consortium of prominent networks - up to 150 sites in US & EU
- Conservative design of Phase III trial to reproduce Phase IIb results
- Phase IIb Trial Results: Randomized, multi-center (16 centers), placebo-controlled, substantial reduction in recurrence rate, peak immunity after 6 months, minimal to no side effects, no SAEs attributable to GP2, led by MD Anderson Cancer Center
- Potential Opportunities to Expand Market:
  - HER2/*neu* 1-2+ patients with Herceptin - increase market from 25% to 75%
  - ✓ Other HLA types – increase from 40-50% up to 80-100% of all patients
  - Combination with CD4/CD8 peptides and checkpoints
  - Other HER2/*neu* cancers
- NASDAQ Ticker "GLSI": Raised $40m since IPO

Greenwich LifeSciences

GLSI Sponsorship of 5k Race in Houston each October

Susan G. Komen race for the cure®

Research

that provides new hope through lifesaving discoveries. Funding breakthroughs that give more time to everyone in their fight against breast cancer.

Care

that ensures all have access to quality screening, diagnosis and treatment for breast cancer. Offering patient support, treatment assistance, even childcare and transportation services.

Community

to support everyone, no matter where they are in their breast cancer journey. Providing a safe place to share, grieve, support and find strength forward.

Action

through advocacy, fighting for government funding and critical patient support and research. We fight for the rights of patients among policymakers and compassionate care for everyone.

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Greenwich LifeSciences
NASDAQ: GLSI

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